CSA GROUP
Laboratory Test Data – UL 9540A

Master Contract: *305512*
Report*: 80239432*
Project: *80239432*

September 12, 2025

Ms. Huijing Ye,
Project Manager
Jiangsu Linyang Energy Storage Technology Co., Ltd
Floor 16-18, Building 1, No.68 Aoti Street
Jianye District, Nanjing
Jiangsu, China

Subject: *Lithium-Ion Rechargeable Battery Rack, model LBR-4-1331.2 /314-L (UL 9540A Test Report)*

Dear *Ms. Huijing Ye*:

We are pleased to inform you that testing of your product per UL 9540A has been completed. Applicable test(s) was witnessed at CCIC-CSA International Certification Co., Ltd. Kunshan Branch – 2F-1, Building C12, No 555 Dujuan Road, Kunshan Economic & Technical Development Zone, Kunshan, Jiangsu. Unit level of test was conducted on the sample provided and the results are enclosed in the test report.

> Note: This Test Report is not an Authorization to apply the CSA Mark to the product. The results contained in the report(s) provided are contingent upon the characteristics of the actual sample(s) used in the investigation. In the absence of a continuing inspection service, CSA provides no assurance, expressed or implied, that the contents of the report are applicable to reproductions of the sample(s). Use or reproduction of the CSA name, logo, or trademark is not permitted without the prior written consent of CSA. No references can be made to this report when using the results of this investigation for the purposes of advertising, promotion or litigation, without the prior written consent of CSA.

Please examine the enclosed documents and contact me if you have any questions or would like us to make any changes.

On behalf of CSA, I would like to thank you for your business and offer our services for your future needs.

Yours truly,

Judy Guo
CSA –*CCIC-CSA International Certification Co., Ltd. Kunshan Branch*
2F-1, Building C12, No 555 Dujuan Road, Kunshan Economic & Technical Development Zone, Kunshan, Jiangsu, China 215331

Encl. [*UL 9540A Test Report*]
Att.1 – Unit charge/discharge conditioning graphs
Att.2 - Photos
Att.3 - Diagram and dimension of test setup
Att.4 - Temperature/voltage graph during testing
Att.5 - Heat release rate graph
Att.6 - Gas generation graphs
Att.7 - Smoke release graph
Att.8 - Heat flux graph
Att.9 - Notable observation during test
Att.10 - Video (separate file)

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Laboratory Test Data – UL 9540A Checklist and Test Result (Unit Level)

ORIGINAL TEST DATA

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Project / Network:	80239432	Description:	Rechargeable Li-ion Battery Rack	

Standard(s): ANSI/CAN/UL 9540A:2019 Fourth Edition, Dated November 12, 2019 - Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems

Testing Laboratory Name:	CCIC-CSA International Certification Co., Ltd. Kunshan Branch
Address:	2F-1, Building C12, No 555 Dujuan Road, Kunshan Economic & Technical Development Zone, Kunshan, Jiangsu, China 215331
Testing Program:	**Custom Test** : Cover Latter ☒, Testing Only ☐

If tests were performed at another facility, then described below:

Testing Laboratory Name:	ChuWeiNeng Testing Technology (Shanghai) Co.Ltd
Address:	Building 3, No.1065 Beihe road, Jiading District, Shanghai
Facility Qualification Number:	N/A

	As above / or describe otherwise
Customer:	Jiangsu Linyang Energy Storage Technology Co., Ltd
Address:	Floor 16-18, Building 1, No.68 Aoti Street, Jianye District, Nanjing, Jiangsu, China

Tested By: Jiaming Huang, Technician
Name, Title

Jiaming Huang — 2025-08-21
Signature — *Date (YYYY-MM-DD)*

☒ **Reviewed by:**
☒ **Witnessed by:** Judy Guo, Certifier
Name, Title

Judy Guo — 2025-08-21
Signature — *Date (YYYY-MM-DD)*

Version6 : 2022-08-02

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Cell Level Test Summary	
Name of test laboratory perform cell level testing:	UL(Changzhou) Quality Technical Service Co., LTD
Unique identification of test report:	Project Number 4791099276
Standard and its edition used for testing:	UL 9540A: 4th edition
Manufacturer:	EVE POWER Co., Ltd.
Brand name / Trademark:	NA
Model number:	MB31
Nominal cell voltage, (V)	3.2
Cell capacity, (Ah)	314
Cell chemistry:	LFP
Physical format of cell:	Prismatic
Approximate dimension, (mm)	(207.2±1)mm by (173.7±1)mm by (71.7±1)mm
Mass, (g)	5600±300g
Method used to initiate thermal runaway:	External heating
Average temperature at which cell first vented excluding gas collection sample, (°C)	154
Average temperature prior to thermal runaway excluding gas collection sample, (°C)	225
Flammable gas generation, (Liter)	--
Total gas generation, (Liter)	192.5
Lower flammability limit (LFL) at ambient temperature (25 ± 5°C), (%)	7.75
Lower flammability limit (LFL) at average gas vent temperature, (%)	7.14
Burning velocity, (Cm/Sec)	75.90
Maximum pressure P_{max}, (psig)	95.40
Gas composition:	See below table

Gas	Measured %
CO	13.775
CO_2	24.315
H_2	49.427
CH_4	5.765
C_2H_2	0.158
C_2H_4	3.583
C_2H_6	1.006
C_3H_6	0.842
C_3H_8	0.295
C4 (Total)	0.538
C5 (Total)	0.129
C6 (Total)	0.038
C_7H_{14}	0.024
$C_3H_6O_3$	0.011
$C_4H_8O_3$	0.093
Total	**100**

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Module Level Test Summary	
Name of test laboratory perform module level testing:	CCIC-CSA International Certification Co., Ltd. Kunshan Branch
Unique identification of test report:	CSA report no. 80239433
Standard and its edition used for testing:	UL 9540A, 5th edition
Manufacturer:	Jiangsu Linyang Energy Storage Technology Co., Ltd
Brand name / Trademark:	N/A
Model number:	LBR-1-332.8/314-L
Nominal voltage rating, (V)	332.8
Nominal capacity rating, (Ah)	314
Approximate dimension, (mm)	LxWxH: (2180±10)mm x (790±10)mm x (252±10)mm
Method used to initiate thermal runaway:	External heating
Number of cells used for initiating thermal runaway:	1
Number of cells exhibited thermal runaway within module:	3
Cell to cell propagation condition:	Yes
Peak chemical heat release rate, (kW)	No flame observed
Flammable gas generation, (Liter)	157.6
Total gas generation, (Liter)	239.7
Weight loss, (%)	0.16
Gas composition:	CO(5.1%)， CO_2(34.3%)， THC(54.6%)， H_2(6.0%)
Additional Information:	N/A

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Master Contract:	305512	Model:	LBR-4-1331.2 /314-L	Page number 5 of 39
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Unit Level Test Summary	
Manufacturer:	Jiangsu Linyang Energy Storage Technology Co., Ltd
Brand name / Trademark:	N/A
Model number:	LBR-4-1331.2 /314-L
Nominal voltage rating, (V)	1331.2
Nominal capacity rating, (Ah)	314
Approximate dimension, (mm)	Battery module(LxWxH): (2180±10)mm x (790±10)mm x (252±10)mm High voltage box(LxWxH): 700mm x700mm x215mm Each battery system includes one high voltage box and four battery modules.
BESS test configuration/intended installation:	Indoor floor mounted non-residential BESS/Container system
Unit certification available?, (Yes/No)	Yes
Standard(s) used to certify product:	UL 1973, 3rd Edition
Certification organization name and its certificate number:	CSA GROUP, Project number:80239428
Electrical configuration of module in BESS:	Cells electrical connection within one module: 104s-1p; modules electrical connection within one unit: 4s-1p
Number of modules in BESS:	One module contains 104 cells and one rack contains 4 modules, one container contains 12 racks
Fire detection and suppression system integral part of BESS: (Yes/No)	Yes, The Fire detection and suppression system is integral with containers but not included in test
Test conducted with fire detection and suppression system: (Yes/No/Not Applicable)	No
Method used to initiate thermal runaway:	External heating
Number of cells used for initiating thermal runaway:	1
Number of cells exhibited thermal runaway within initiating module:	3
Number of modules exhibited thermal runaway within initiating BESS:	1

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Master Contract:	305512	Model:	LBR-4-1331.2 /314-L	
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Unit Level Test Summary	
Cell to cell propagation condition:	Yes, initiating cell went into thermal runaway and propagated to one adjacent cell
Peak chemical heat release rate, (kW)	3.79(No flame observed)
Peak convective heat release Rate, (kW)	0
Flammable gas generation, (Liter)	Before Flaming: 474.9 After Flaming: --
Total gas generation, (Liter)	Before Flaming: 593.5 After Flaming: --
Gas composition:	Before Flaming: CO(12.4%)，CO_2(20.0%)，THC(48.7%)，H_2(18.9%) After Flaming: --
Maximum wall surface temperature, (°C)	58.9
Maximum target BESS temperature, (°C)	34.8
Maximum incident heat flux on target wall surfaces, (kW/m^2)	1.31
Maximum incident heat flux on target BESS, (kW/m^2)	0.1
Maximum incident heat flux of egress path, (kW/m^2)	N/A
Total smoke release, (m^2)	152.43
Peak smoke release rate, (m^2/s)	0.9452
Additional Information:	N/A

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Performance Unit Level Test (Non – Residential Indoor Floor Mounted)		
Requirement	**Comments**	**Verdict**
Flaming outside the initiating BESS unit is not observed	No flaming outside the initiating BESS unit was observed	P
Surface temperatures of modules within the target BESS units adjacent to the initiating BESS unit do not exceed the temperature at which thermally initiated cell venting occurs	The maximum surface temperature of modules within the target BESS units adjacent to the initiating BESS unit is 34.8°C and do not exceed the temperature at which thermally initiated cell venting occurs	P
For BESS units intended for installation in locations with combustible constructions, surface temperature measurements on wall surfaces do not exceed 97°C (175°F) of temperature rise above ambient	BESS will not be installed in locations with combustible constructions	N/A
Explosion hazards are not observed, including deflagration, detonation or accumulation (to within the flammability limits in an amount that can cause a deflagration) of battery vent gases	No explosion hazards were observed	P
Heat flux in the center of the accessible means of egress shall not exceed 1.3 kW/m^2	Heat flux in the center of the accessible means of egress was not exceed 1.3 kW/m2	P
Summary of Result: No further testing is required based on above performance condition indicated in Section 9.8 of UL 9540A 4th Edition.		

Possible test case verdicts:

- Test object does not apply to the test object: N/A
- Test object does meet the requirement: P (Pass)
- Test object does not meet the requirement: F (Fail)
- Test object waived based construction detail: W (Waived)

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Clause	Requirement + Test	Result - Remark	Verdict
	Construction		
5	**General**	---	---
5.3	**Battery energy storage system unit**	---	---
5.3.1	BESS/Battery system certification available? (Yes/No)	Yes	---
	Standard(s) used to certify product:	UL 1973, 3rd Edition	---
5.3.2	BESS/Battery system component documentation	☒ BESS/Battery system certification was available – Component detail not documented. ☐ BESS/Battery system certification was not available – See list of critical components in attachment section. ☐ Other(explain):	---
	BESS/Battery system enclosure approximate dimension, (mm)	Battery module(LxWxH): (2180±10)mm x (790±10)mm x (252±10)mm High voltage box(LxWxH): 700mm x700mm x215mm Each battery system includes one high voltage box and four battery modules. Container (LxWxH): 6058mmx2438 mmx2896mm	---
	BESS/Battery system enclosure material:	Container: metallic	---
	Based on configuration of BESS, test conducted on:	☐ BESS ☒ Battery system	---
5.3.3	Fire detection system	☐ Integral part of DUT, test conducted with fire detection system. ☒ Integral part of DUT, test conducted without fire detection system. ☐ Not integral part of DUT	---
	Fire suppression system	☐ Integral part of DUT, test conducted with fire suppression system. ☒ Integral part of DUT, test conducted without fire suppression system. ☐ Not integral part of DUT	---
5.3.4	Unit level test report	See below	---
	Performance		
9	**Unit level**	---	---
9.1	**Sample and test configuration**	---	---

Clause	Requirement + Test	Result - Remark	Verdict
9.1.1	The unit level test was conducted with BESS units installed as described in the manufacturer's instructions and this section.	Confirmed	P
	BESS test configuration:	This DUT is a container system BESS intended for outdoor installation, and choose one battery system(one high voltage box and four battery modules install in battery rack) as initial unit to test in accordance with the indoor floor mounted	---
9.1.2	Unit level test was conducted in which internal fire condition created as per module level test.	Confirmed	P
	Test setup include initiating BESS unit and target BESS unit representative of an installation.	Refer to Attachment 3 Unit Level Test Setup for the detail of the test configuration layout.	P
	Additional representative test configuration based on test configuration.	---	---
	Separation distances between initiating and target units were representative of the installation.	Confirmed, refers to Attachment 3 Unit Level Test Setup for details.	P
	Testing conducted outdoor for BESS intended for outdoor installation only. Following controls and environmental conditions were in place.	Test conducted per indoor floor mounted non-residential application	N/A
	a) Wind screens were utilized with a maximum wind speed maintained at ≤ 12 mph	---	N/A
	b) Temperature range was within 10°C to 40°C	---	N/A
	c) The humidity was < 90% RH	---	N/A
	d) There was sufficient light to observe the testing;	---	N/A
	e) There was no precipitation during the testing;	---	N/A
	f) There was control of vegetation and combustibles in the test area to prevent any impact on the testing and to prevent inadvertent fire spread from the test area; and	---	N/A

Clause	Requirement + Test	Result - Remark	Verdict
	g) There were protection mechanisms in place to prevent inadvertent access by unauthorized persons in the test area and to prevent exposure of persons to any hazards as a result of testing.	---	N/A
9.1.2.1	For a container system BESS including those intended for outdoor installation only, the unit level test performed in accordance with the indoor floor mounted unit level test using the battery system racks as the test units and with the test installation set up in accordance with the installation layout within the container.	Confirmed	P
9.1.3	Based on configuration and design of BESS, test conducted on:	☐ BESS ☒ Battery system	---
9.1.4	Initiating BESS unit contain components representative of a BESS unit in a complete installation.	Confirmed	P
	Combustible components that interconnect the initiating and target BESS units were included.	Confirmed	P
9.1.5	Target BESS units include the outer cabinet (if part of the design), racking, module enclosures, and components that retain cells components.	Confirmed	P
	The target BESS unit module enclosures did not contain cells.	Confirmed	P
9.1.6	Initiating BESS unit was at the maximum operating state of charge (MOSOC).	Confirmed	P
	After charging and prior to testing, the initiating BESS was rested for a maximum period of 8 h at room ambient.	See table 2 for details	P
9.1.7	BESS unit test conducted as per following condition.	---	P
	a) Integral fire suppression system provided with the DUT.	--	N/A
	b) Without Integral fire suppression system.	Integral part of BESS, test conducted on battery system without fire suppression system.	P
9.1.8	Electronic and software control were not relied upon for this testing.	Confirmed	P
	BESS unit test conducted with Integral fire suppression system meet UL 840 and considered reliable for this testing.	Test conducted without fire suppression system	N/A

Clause	Requirement + Test	Result - Remark	Verdict
9.2	**Test method – Indoor floor mounted BESS units**	---	---
9.2.1	Test room environment was controlled to prevent drafts that may affect test results.	Confirmed	P
	At the start of the test, the room ambient temperature was not less than 10°C (50°F) nor more than 32°C (90°F).	See table 2 for details	P
	Ambient temperature range during test, °C	See table 2 for details	P
9.2.2	Any access door(s) or panels were closed, latched and locked at the beginning and duration of the test.	Test conducted on battery system without access door(s) or panels.	N/A
9.2.3	The initiating BESS unit was positioned adjacent to two instrumented wall sections.	Confirmed	P
9.2.4	Instrumented wall sections were extended not less than 0.49 m (1.6 ft) horizontally beyond the exterior of the target BESS units.	Confirmed	P
9.2.5	Instrumented wall sections were at least 0.61-m (2-ft) taller than the BESS unit height, but not less than 3.66 m (12 ft) in height above the bottom surface of the unit.	Confirmed	P
9.2.6	The surface of the instrumented wall sections was covered with 16-mm (5/8-in) gypsum wall board and painted flat black.	Confirmed	P
9.2.7	The initiating BESS unit was centered underneath an appropriately sized smoke collection hood of an oxygen consumption calorimeter.	Confirmed	P
9.2.8	The light transmission in the calorimeter's exhaust duct was measured.	Confirmed	P
	White light source and photo detector was used for the duration of the test.	See attachment 7 for details	P
	Smoke release rate was calculated as per following formula. $SRR = 2.303\left(\frac{V}{D}\right) Log_{10}\left(\frac{I_o}{I}\right)$	Confirmed	P
9.2.9	The chemical and convective heat release rates were measured for the duration of the test.	See attachment 5 for details	P
	Chemical heat release rate was calculated as per following formula.	Confirmed	P

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Clause	Requirement + Test	Result - Remark	Verdict
	$HRR_1 = \left[E \times \varphi - (E_{co} - E) \times \frac{1-\varphi}{2} \times \frac{X_{co}}{X_{O_2}}\right] \times \frac{\dot{m}_e}{1+\varphi \times (\alpha - 1)} \times \frac{M_{O_2}}{M_a} \times (1 - X^o_{H_2O}) \times X^o_{O_2}$		
9.2.10	The heat release rate measurement system shall be calibrated using an atomized heptane diffusion burner.	Confirmed	P
9.2.11	The convective heat release rate was measured during test.	See attachment 5 for details	P
	Thermopile, a velocity probe, and a Type K thermocouple, located in the exhaust system of the exhaust duct were used for measurement.	Confirmed	P
9.2.12	Convective heat release rate was calculated as per following formula. $HRR_c = V_e A \frac{353.22}{T_e} \int_{T_o}^{T} C_p dT$	Confirmed	P
9.2.13	Physical spacing between BESS units (both initiating and target) and adjacent walls were representative of the intended installation.	See Attachment 3 for details	P
9.2.14	Separation distances was specified by the manufacturer for distance between:	See Attachment 3 for details	P
	a) The BESS units and the instrumented wall sections.	See Attachment 3 for details	P
	b) Adjacent BESS units.	See Attachment 3 for details	P
9.2.15	Wall surface temperature measurements was collected for BESS intended for installation in locations with combustible construction.	Confirmed	P
9.2.16	Wall surface temperatures was measured in vertical array(s) at 152-mm (6-in) intervals for the full height of the instrumented wall sections.	Confirmed	P
	No. 24-gauge or smaller, Type-K exposed junction thermocouples were used for measurement.	Confirmed	P
	The thermocouples were placed horizontally positioned in the wall locations anticipated to receive the greatest thermal exposure.	Confirmed	P

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Clause	Requirement + Test	Result - Remark	Verdict
	Temperatures was measured continuously, averaging over every 60 second interval.	Confirmed	P
	The maximum of these averages was documented for each thermocouple location.	Confirmed	P
9.2.17	Thermocouples were secured to gypsum surfaces by the use of staples placed over the insulated portion of the wires.	Confirmed	P
	The thermocouple tip was depressed into the gypsum so as to be flush with the gypsum surface at the point of measurement and held in thermal contact with the surface at that point by the use of pressure-sensitive paper tape.	Confirmed	P
9.2.18	Heat flux was measured with the sensing element of at least two water-cooled Schmidt- Boelter or Gardon gauges at the surface of each instrumented wall.	Confirmed	P
	a) Both were collinear with the vertical thermocouple array.	Confirmed	P
	b) One was positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module	Confirmed	P
	c) One was positioned at the elevation estimated to receive the greatest heat flux during potential propagation of thermal runaway within the initiating BESS unit.	Confirmed	P
	Heat flux was measured continuously, averaging over every 60 second interval.	Confirmed	P
	The maximum of these averages was documented for each gauge location.	Confirmed	P
9.2.18.1	Heat flux measurements on walls were waived for residential units that are tested with the cheesecloth indicator.	Non-residential application	N/A
9.2.18.2	With reference to 9.2.18, if b) and c) were deemed to be at the same location, only one gauge was installed on the wall for the measurement.	Non-residential application	N/A
9.2.19	Heat flux was measured with the sensing element of at least two water-cooled Schmidt- Boelter or Gardon gauges at the	Confirmed	P

Clause	Requirement + Test	Result - Remark	Verdict
	surface of each adjacent target BESS unit that faces the initiating BESS unit:		
	a) One was positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module within the initiating BESS	Confirmed	P
	b) One was positioned at the elevation estimated to receive the greatest surface heat flux due to the thermal runaway of the initiating BESS.	Confirmed	P
	Heat flux was measured continuously, averaging over every 60 second interval.	Confirmed	P
	The maximum of these averages was documented for each gauge location.	Confirmed	P
9.2.19.1	Heat flux measurements on target units were waived for residential units that are tested with the cheesecloth indicator.	Non-residential application	N/A
9.2.19.2	With reference to 9.2.19, if a) and b) were deemed to be at the same location, only one gauge was installed on the target unit for the measurement.	Confirmed	P
9.2.20	For non-residential use BESS, heat flux was measured with the sensing element of at least one water-cooled Schmidt-Boelter or Gardon gauge positioned at one for the following location.	Confirmed	P
	a. At the mid height of the initiating unit in the center of the accessible means of egress.	See attachment 8 for details	P
	b. At the point where the majority of off-gas venting was expected from the initiating unit in the center of the accessible means of egress.	See attachment 8 for details	P
9.2.21	No. 24-gauge or smaller, Type-K exposed junction thermocouples was installed to measure the temperature of the surface proximate to the cells and between the cells and exposed face of the initiating module.	Confirmed	P
	Each non-initiating module enclosure within the initiating BESS unit was instrumented with at least one No. 24-gauge or smaller Type-K thermocouple(s) to provide data to monitor the thermal conditions within non-initiating modules.	Confirmed	P

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Clause	Requirement + Test	Result - Remark	Verdict
	Additional thermocouples shall be placed to account for convoluted enclosure interior geometries.	Confirmed	P
	Temperatures was measured continuously, averaging over every 60 second interval.	Confirmed	P
	The maximum of these averages was documented for each thermocouple location.	Confirmed	P
9.2.22	For residential use BESS, the DUT was covered with a single layer of cheese cloth ignition indicator.	Non-residential application	N/A
	The cheesecloth was untreated cotton cloth running 26 – 28 m^2/kg with a count of 28 – 32 threads in either direction within a 6.45 cm^2 (1 in^2) area.	Non-residential application	N/A
9.2.23	An internal fire condition in accordance with the module level test was created within a single module in the initiating BESS unit.	Confirmed	P
	a) The position of the module was selected to present the greatest thermal exposure to adjacent modules (e.g. above, below, laterally), based on the results from the module level test;	Confirmed	P
	b) The setup (i.e. type, quantity and positioning) of equipment for initiating thermal runaway in the module was same as that used to initiate and propagate thermal runaway within the module level test.	Confirmed	P
9.2.24	The composition, velocity and temperature of the initiating BESS unit vent gases was measured within the calorimeter's exhaust duct.	Confirmed	P
	The hydrocarbon content of the vent gas was measured using flame ionization detection.	Confirmed	P
	Hydrogen gas was measured with a palladium-nickel thin-film solid state sensor.	Confirmed	P
	Composition, velocity and temperature instrumentation were collocated with heat release rate calorimetry instrumentation.	Confirmed	P

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Clause	Requirement + Test	Result - Remark	Verdict
9.2.25	The hydrocarbon content of the vent gas was additionally measured a Fourier-Transform Infrared Spectrometer with a minimum resolution of 1 cm^{-1} and a path length of at least 2.0 m (6.6 ft), or equivalent gas analyzer.	FTIR analysis was not used in accordance with the Certification Requirement Decision: Corrections to gas measurement methods to make FTIR as an option for measuring hydrocarbon contents of gas emissions and to include Hydrogen measurements during the Unit Level Test. FTIR was considered redundant to the other gas measurement methods used.	N/A
9.2.26	The test was terminated at:	See below	P
	a) Temperatures measured inside each module within the initiating BESS unit return to ambient temperature;	Confirmed	P
	b) The fire propagates to adjacent units or to adjacent walls; or	No fire propagates to adjacent units or to adjacent walls	N/A
	c) A condition hazardous to test staff or the test facility requires mitigation.	No condition hazardous to test staff or the test facility	N/A
9.2.27	For residential use systems, the gas collection data gathered was compared to the smallest room installation specified by the manufacturer to determine if the flammable gas collected exceeds 25% LFL in air.	Non-residential use systems.	N/A
9.3	**Test method – Outdoor ground mounted units**	Test conducted followed the requirements of indoor floor mounted non-residential application	N/A
9.4	**Test Method – Indoor wall mounted units**	Test conducted followed the requirements of indoor floor mounted non-residential application	N/A
9.5	**Test Method – Outdoor wall mounted units**	Test conducted followed the requirements of indoor floor mounted non-residential application	N/A
9.6	**Rooftop and open garage installations**	Test conducted followed the requirements of indoor floor mounted non-residential application	N/A
9.7	**Unit level test report**	See below	P
9.7.1	Type of installation considered during unit level testing:	Indoor floor mounted non-residential application	P
9.7.2	Additional installation represented by type of installation considered during unit level testing:	Container system installation	P

Clause	Requirement + Test	Result - Remark	Verdict
9.7.3	Unit level report include following information.	See below	P
	a) Unit manufacturer name and model number (and whether UL 9540 compliant);	Jiangsu Linyang Energy Storage Technology Co., Ltd. Battery system model no. LBR-4-1331.2 /314-L UL 9540 complaint, CSA Group, project number: 80239430	P
	b) Number of modules in the initiating BESS unit;	4	P
	c) The construction of the initiating BESS unit per 5.3;	Confirmed	P
	d) Fire protection features / detection / suppression systems within unit;	Test conducted without fire detection / suppression system.	P
	e) Module voltage(s) corresponding to the tested SOC;	Confirmed, see table 3 for details	P
	f) The thermal runaway initiation method used;	External heating	P
	g) Location of the initiating module within the BESS unit;	Refer to TC map of initiating BESS and target BESS in Attachment 2	P
	h) Diagram and dimensions of the test setup including mounting location of the initiating and target BESS units, and the locations of walls, ceilings, and soffits;	Refer to TC map of initiating BESS and target BESS in Attachment 2	P
	i) Observation of any flaming outside the initiating BESS enclosure and the maximum flame extension;	No flame observed	N/A
	j) Chemical and convective heat release rate versus time data;	See attachment 5 for details	P
	k) Separation distances from the initiating BESS unit to target walls;	See attachment 3 for details	P
	l) Separation distances from the initiating BESS unit to target BESS units;	See attachment 3 for details	P
	m) The maximum wall surface and target BESS temperatures achieved during the test and the location of the measuring thermocouple;	See attachment 5 for details	P
	n) The maximum ceiling or soffit surface temperatures achieved during the indoor or outdoor wall mounted test and the location of the measuring thermocouple;	Indoor floor mounted non-residential applications, no ceiling or soffit surface considered when testing	N/A
	o) The maximum incident heat flux on target wall surfaces and target BESS units;	See attachment 8 for details	P

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Clause	Requirement + Test	Result - Remark	Verdict
	p) The maximum incident heat flux on target ceiling or soffit surfaces achieved during the indoor or outdoor wall mounted test;	Indoor floor mounted non-residential Applications.	N/A
	q) Gas generation and composition data;	See attachment 6 for details	P
	r) Peak smoke release rate and total smoke release data;	See attachment 7 for details	P
	s) Indication of the activation of integral fire protection systems and if activated the time into the test at which activation occurred;	Not applicable	N/A
	t) Observation of flying debris or explosive discharge of gases;	Not observed	P
	u) Observation of re-ignition(s) from thermal runaway events;	Not observed	P
	v) Observation(s) of sparks, electrical arcs, or other electrical events;	Not observed	P
	w) Observations of the damage to: 1) The initiating BESS unit; 2) Target BESS units; 3) Adjacent walls, ceilings, or soffits	Not observed	P
	x) Photos and video of the test.	See attachments 2 and 10 for details	P

Table 1 – Module charge/discharge specification			
Charging method	CP	Discharging method	CP
Charge current, (kW)	52.2	Discharge current, (kW)	52.2
Charge voltage, (Vdc)	---	---	---
Charge end voltage, (Vdc)	379.6V for module/3.65V for cell	Discharge end voltage, (Vdc)	280.8V for module/2.7V for cell
Manufacturer recommended charge temperature, (°C)	0 to 50	Manufacturer recommended discharge temperature, (°C)	-20 to 55
Note: The charge/discharge was done on battery module one by one according to above parameters, and all modules were charged to 100% SOC as manufacture specification before test.			

Table 2 – Unit rest duration				
Sample Number	Final charge end time		Test start time	
	Date (YYYY-MM-DD)	Time (HH:MM AM/PM)	Date (YYYY-MM-DD)	Time (HH:MM AM/PM)
CWN250815XF03	2025-08-21	8:28 AM	2025-08-21	14:45 PM
Ambient temperature during unit conditioning				
Ambient Lab Temperature, (°C)		Relative Humidity, (%RH)		
20.0 to 30.0		25.0 to 75.0		

Table 3 – Unit level test	
Sample Number:	CWN250815XF03
Ambient temperature at start of test, (°C)	27.9
Ambient temperature range during test, (°C)	22.2~28.3
Relative humidity, (%RH)	71
Number of cells used for initiating thermal runaway:	1
Open circuit voltage before test, (Vdc)	See attachment 2 for details
External film heater ramp rate, (°C/min)	4.5
Other method used to initiate thermal runaway:	N/A
Location of cell and module for initiating thermal runaway:	See attachment 2 for details
Number of cells exhibited thermal runaway within initiating module:	3
Number of modules exhibited thermal runaway within initiating BESS:	1
Location of cell and module exhibited thermal runaway within initiating BESS:	See attachment 2 for details

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Table 3 – Unit level test	
Cell to cell propagation condition:	Yes, initiating cell went into thermal runaway and propagated to one adjacent cell
Peak chemical heat release rate, (kW)	3.79 (No flame observed)
Peak convective heat release rate, (kW)	0
Flammable gas generation, (Liter)	Before Flaming: 474.9 After Flaming: --
Total gas generation, (Liter)	Before Flaming: 593.5 After Flaming: --
Peak smoke release rate, (m^2/sec)	0.9452
Total smoke release rate, (m^2)	152.43

Table 4 – Gas composition			
Gas Component		**Volume Released (Before Flaming) (Liter)**	**Volume Released (After Flaming) (Liter)**
Carbon Monoxide	CO	73.8	---
Carbon Dioxide	CO_2	118.6	---
Hydrogen (Palladium nickel sensor)	H_2	112.4	---
Total Hydrocarbons (equivalent to CH4, measured by FID)	---	288.7	---

Table 5 – Critical observation	
Condition	**Comment**
Any flaming outside the initiating BESS enclosure and the maximum flame extension:	Not observed
Flying debris	Not observed
Explosive discharge of gases	Not observed
Re-ignition(s) from thermal runaway events	Not observed
Sparks	Not observed
Electrical arcs	Not observed
Other electrical events	N/A
Damage to the initiating BESS unit	Not observed
Damage to target BESS units;	Not observed
Damage to adjacent walls	Not observed
Damage to ceilings	N/A
Damage to soffits	N/A

Attachments

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Attachment 1 - Unit charge/discharge conditioning graphs

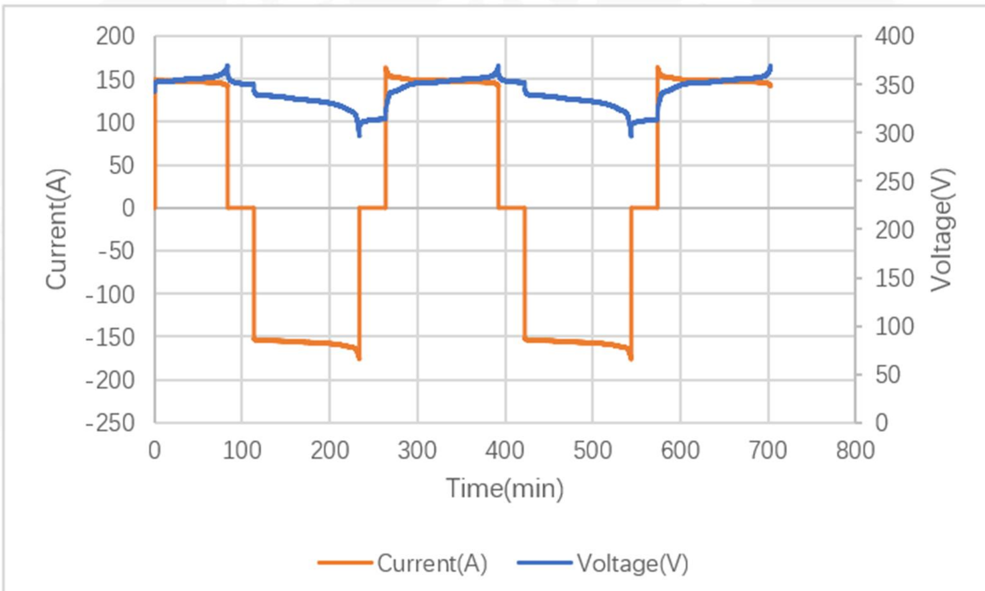

Figure 1: initial module charge/discharge profile

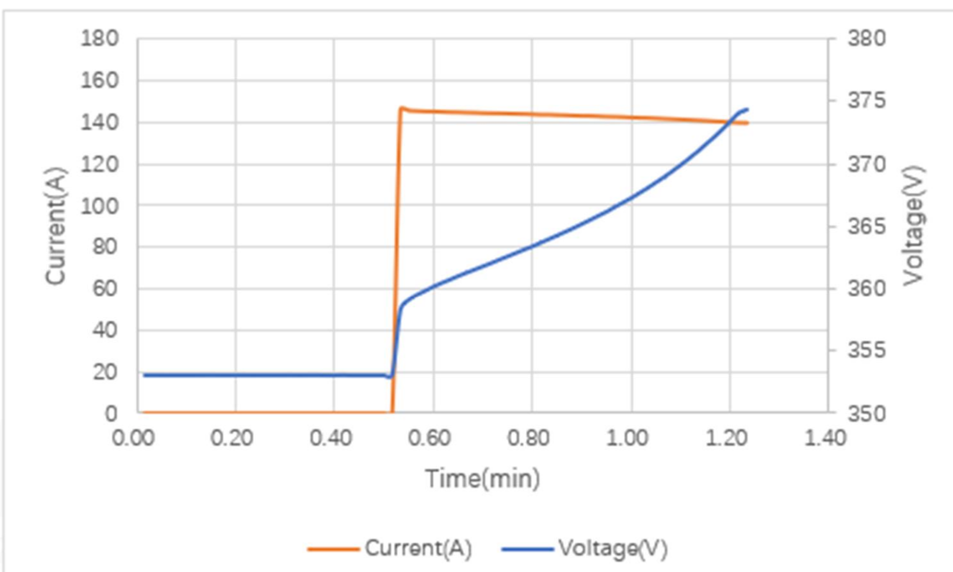

Figure 2: Module 1 charge/discharge profile

Attachment 1 - Unit charge/discharge conditioning graphs

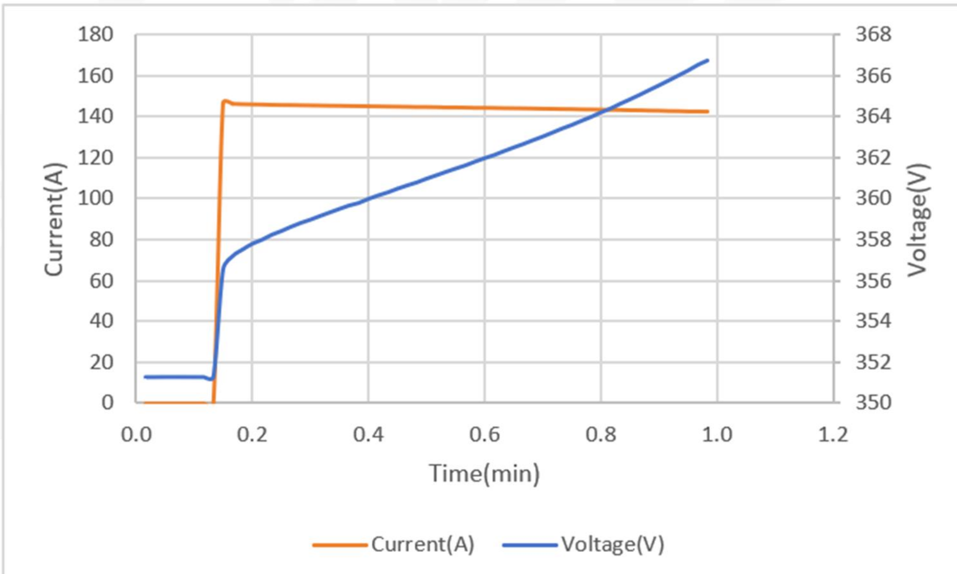

Figure 3: Module 2 charge/discharge profile

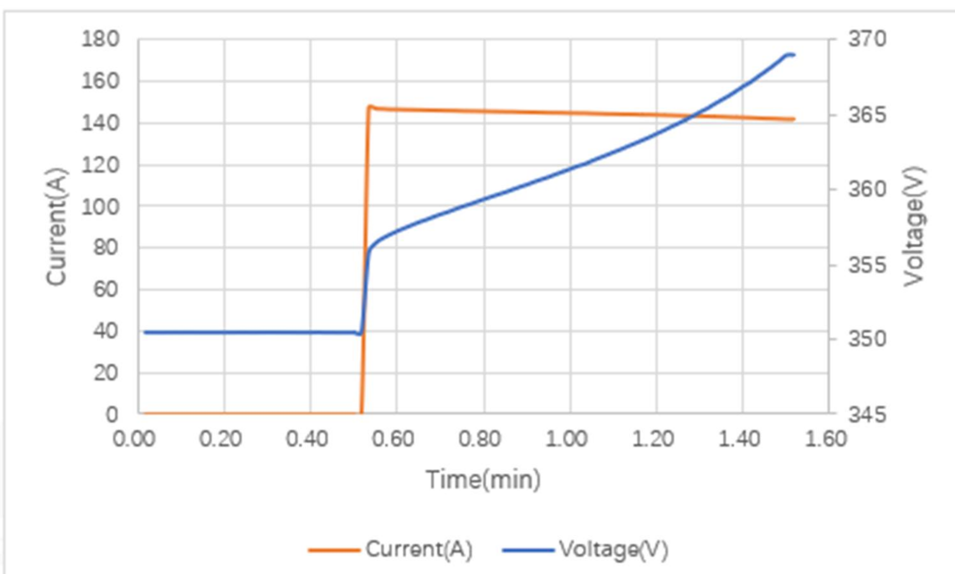

Figure 4: Module 4 charge/discharge profile

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Attachment 2 – Photos

General sample photos

Figure 1: Initiating module top view

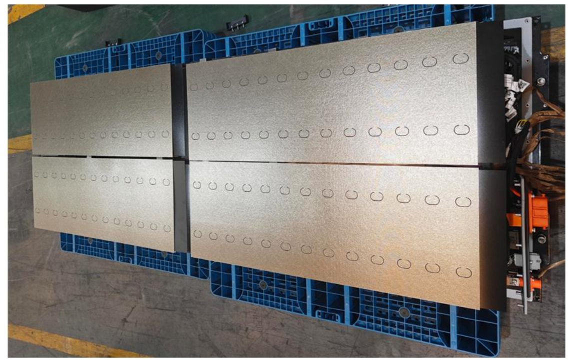

Figure 2: Initiating module removing top cover

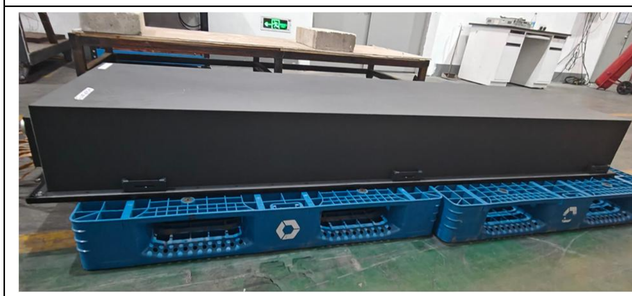

Figure 3: Initiating module side view

Figure 4: Initiating module front view

Figure 5: Rack front view

Figure 6: Rack side view

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Attachment 2 – Photos

Photos with heater and thermocouple installation

Figure 7: Initiating module layout

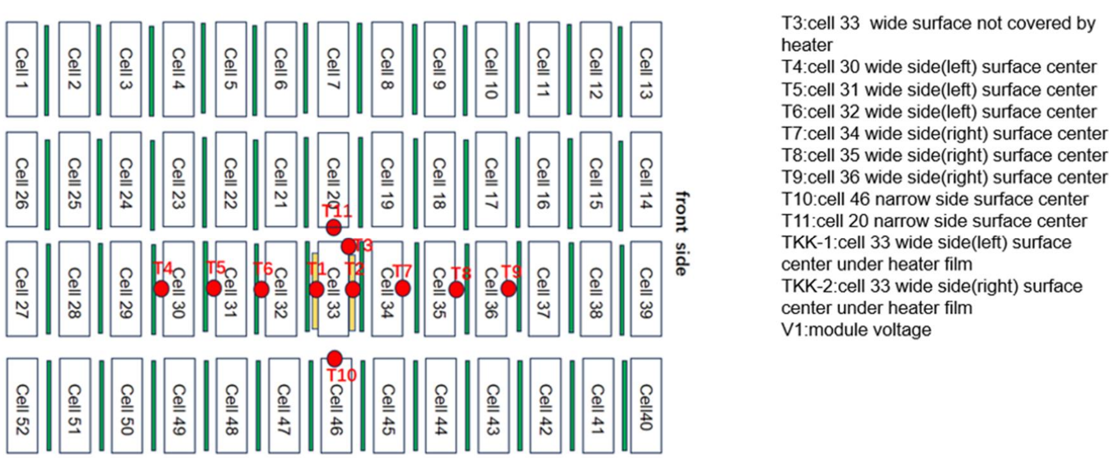

Figure 8: Initiating module thermocouple and voltage layout

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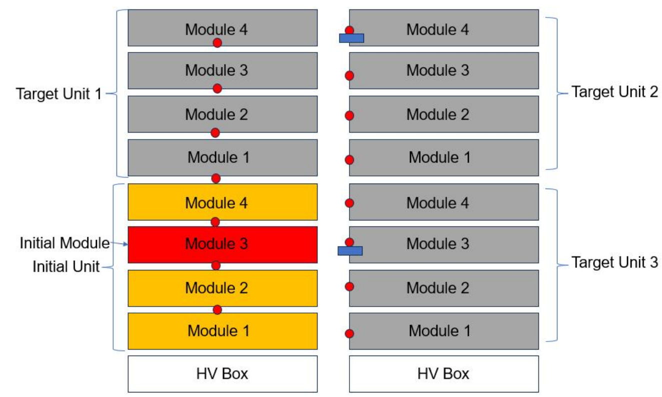
Attachment 2 – Photos



Item	Description
TA50	Module 1 enclosure top surface(Initial Unit)
TA51	Module 2 enclosure top surface(Initial Unit)
TA52	Module 4 enclosure bottom surface(Initial Unit)
TA53	Module 1 enclosure bottom surface(Target Unit 1)
TA54	Module 2 enclosure bottom surface(Target Unit 1)
TA55	Module 3 enclosure bottom surface(Target Unit 1)
TA56	Module 4 enclosure bottom surface(Target Unit 1)
TA57	Module 1(Target Unit 2) enclosure left side surface, face to initial module vent valve
TA58	Module 2(Target Unit 2) enclosure left side surface, face to initial module vent valve
TA59	Module 3(Target Unit 2) enclosure left side surface, face to initial module vent valve
TA60	Module 4(Target Unit 2) enclosure left side surface, face to initial module vent valve
TA61	Module 1(Target Unit 3) enclosure left side surface, face to initial module vent valve
TA62	Module 2(Target Unit 3) enclosure left side surface, face to initial module vent valve
TA63	Module 3(Target Unit 3) enclosure left side surface, face to initial module vent valve
TA64	Module 4(Target Unit 3) enclosure left side surface, face to initial module vent valve

● :Thermalcouple; ▬ :Heat flux

Figure 9: Rack thermocouple and heat flux lay out

Photos during test in progress

Figure 10: At test start (Time in 14:45)

Figure 11: During cell venting (Time in 15:27)

Figure 12: During thermal runaway (Time in 15:45)

Figure 13: During thermal runaway (Time in 15:49)

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Attachment 2 – Photos

Figure 14: During thermal runaway (Time in 15:59)

Figure 15: Test end (Time in 8:00 next day)

Photos after test

Figure 16: Rack front view

Figure 17: Rack side view

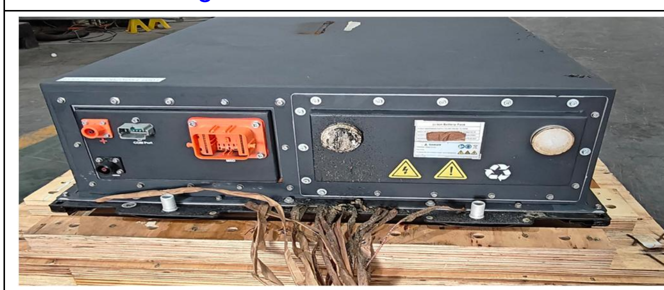

Figure 18: Initiating module front view

Figure 19: Initiating module top view

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Attachment 3 - Diagram and dimension of test setup

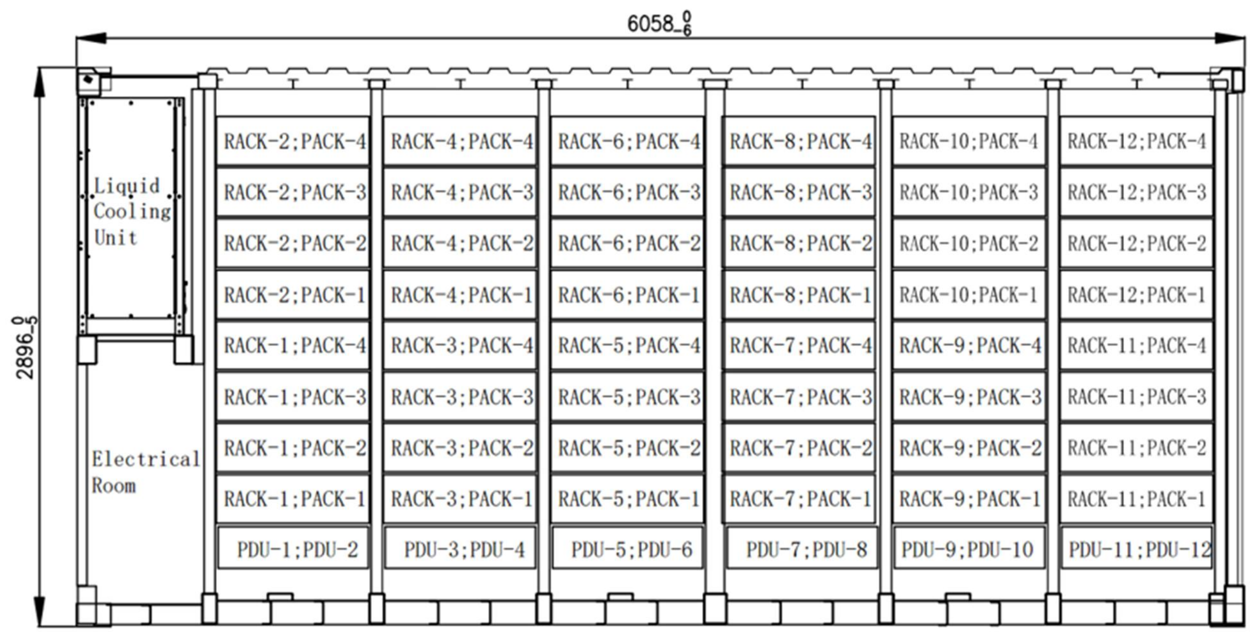

Figure 1 Container internal layout

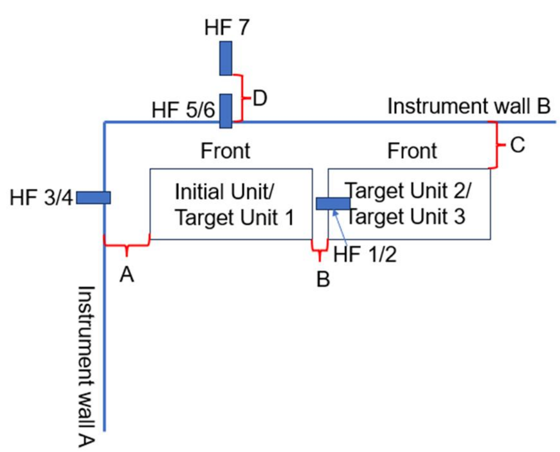

Figure 2 Test set up layout

Separation distance and other critical dimension detail		
Location	Required by manufacturer (mm)	Measured (mm)
The distance between instruction wall A and initial unit	60	60
The distance between target unit 2 and initial unit	60	60

Attachment 3 - Diagram and dimension of test setup

The distance between instrument wall B and initial unit	50	50
The distance between instrument wall B and egress path center	1000	1000

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Attachment 4 - Temperature/voltage graph during testing

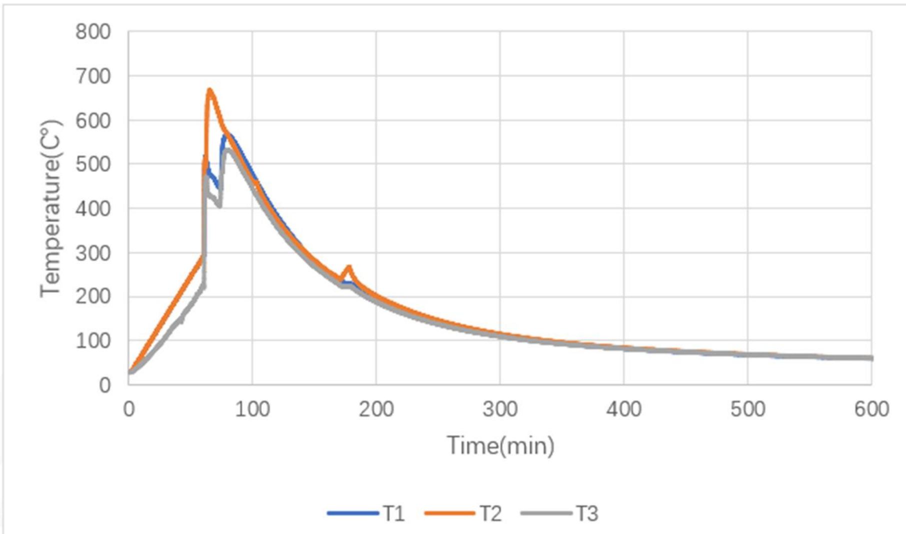

Figure 1: Initial cell surface temperature within initiating module

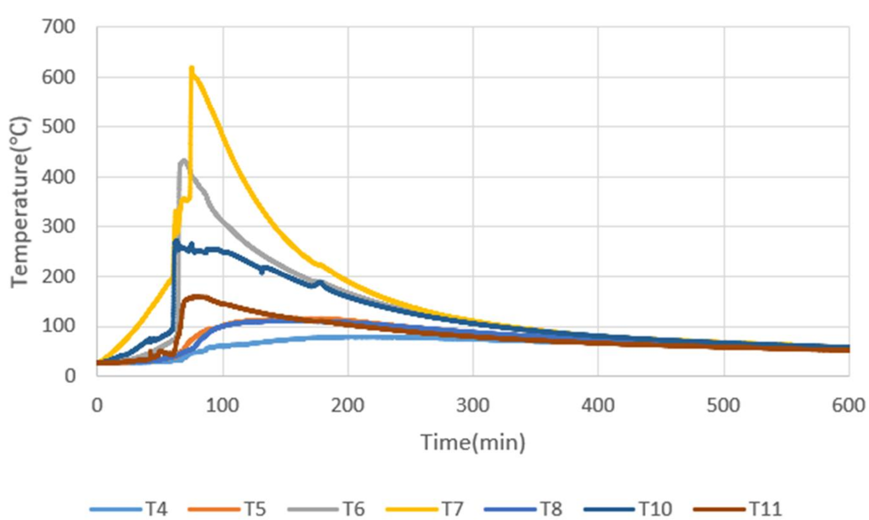

Figure 2: Cell surface temperature within initiating module

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Attachment 4 - Temperature/voltage graph during testing

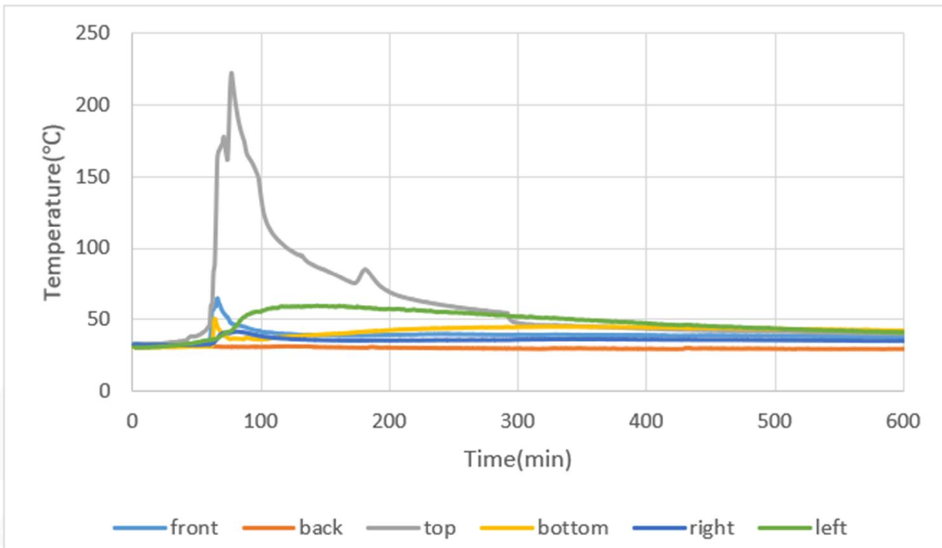

Figure 3 : Initial module enclosure surface temperature

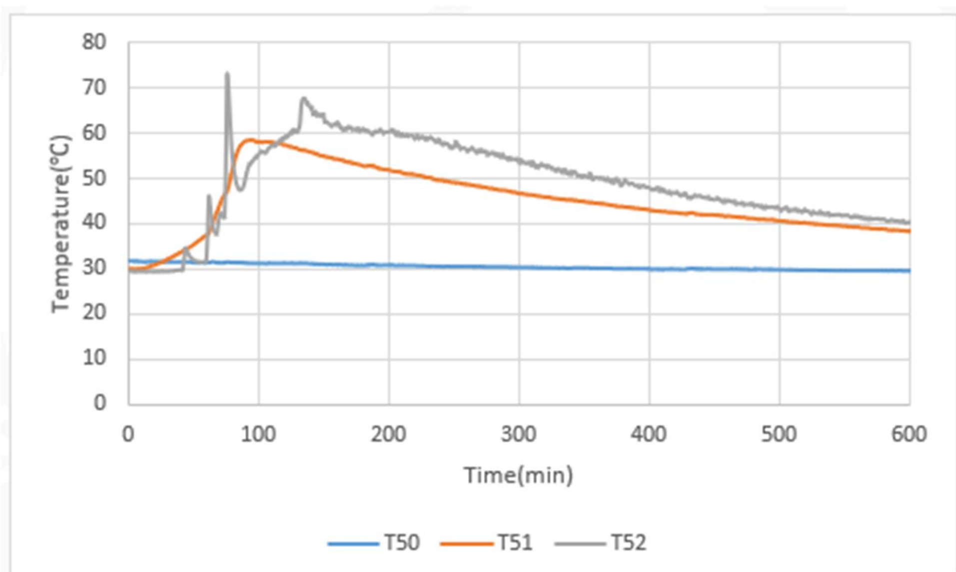

Figure 4: Module enclosure surface temperature within initial unit

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Attachment 4 - Temperature/voltage graph during testing

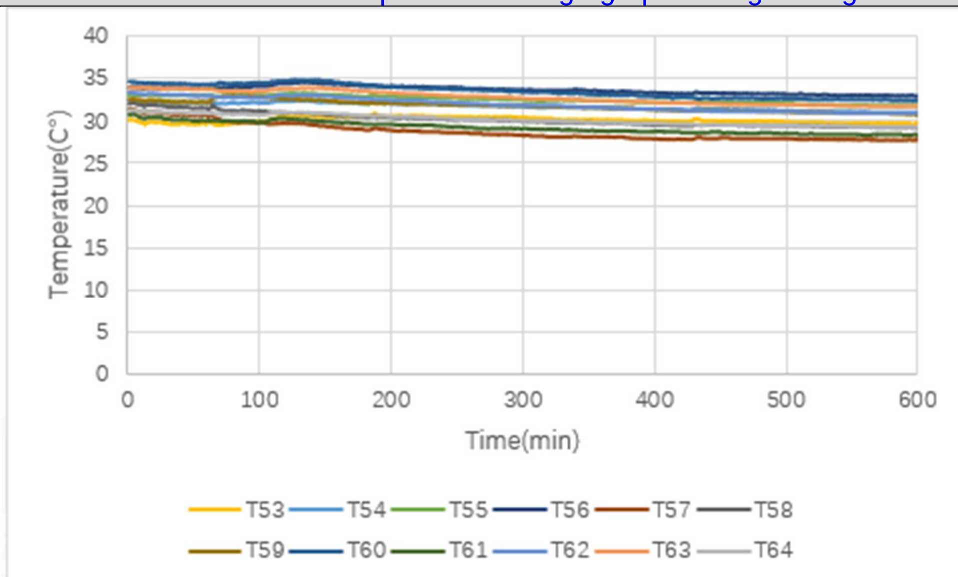

Figure 5: Module enclosure surface temperature within target unit

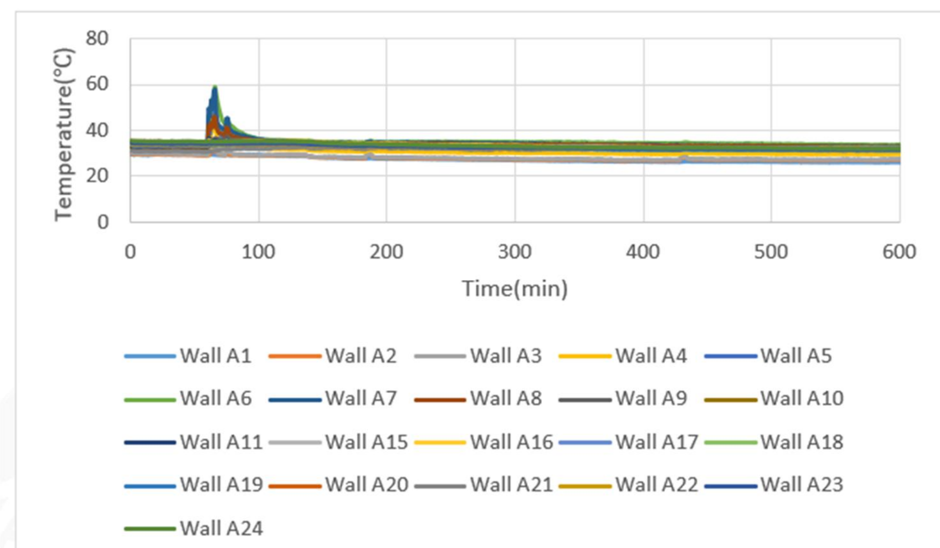

Figure 6: Instruction wall A surface temperature

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Attachment 4 - Temperature/voltage graph during testing

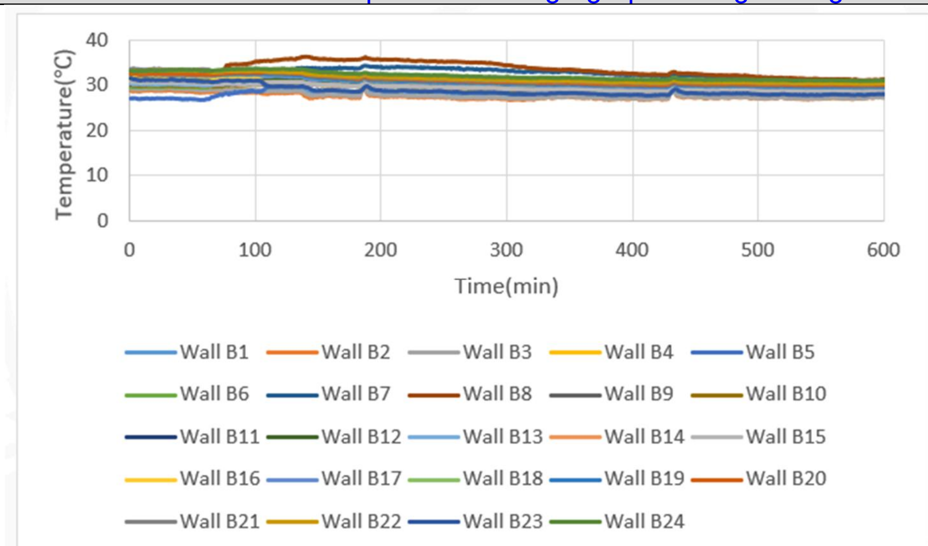

Figure 7: Instruction wall B surface temperature

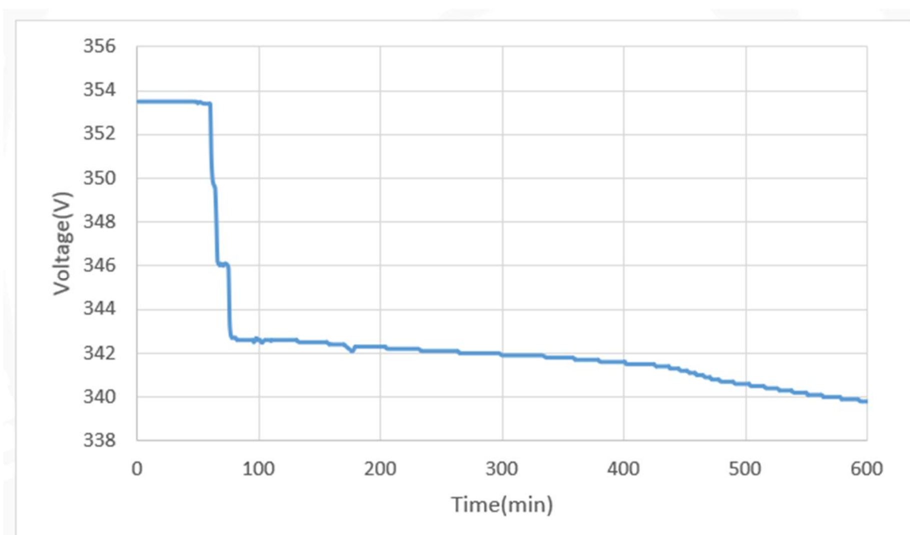

Figure 8: Initial module voltage

Maximum temperature measurement		
Location	**Temperature limit (°C)**	**Measured maximum temperature (°C)**
Maximum Temperature at initiating unit enclosure	N/A	222.4
Maximum Temperature at target unit	154	72.8
Maximum Temperature at Instruction wall A	125.3	58.9
Maximum Temperature at Instruction wall B	125.3	36.4

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Attachment 4 - Temperature/voltage graph during testing

Open circuit voltage measurement		
Location	**Before testing (Vdc)**	**After testing (Vdc)**
Module 1 in initiating unit	358.3	357
Module 2 in initiating unit	358.7	357
Module 3 (initial module) in initiating unit	353.9	336
Module 4 in initiating unit	355.2	355

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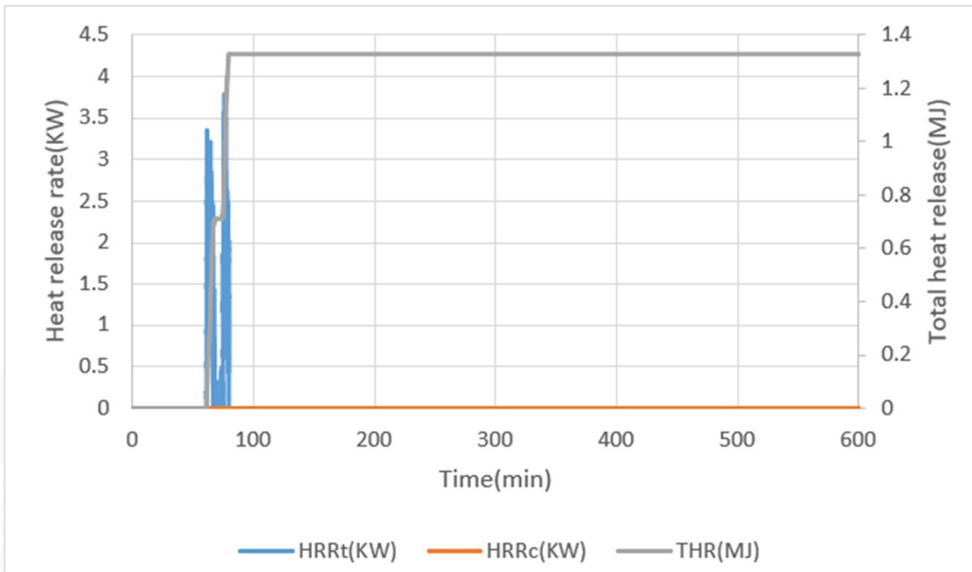
Attachment 5 - Heat release rate graphs



Note: No flame observed during test, so the chemical heat release rate and total heat release are identified as noise.

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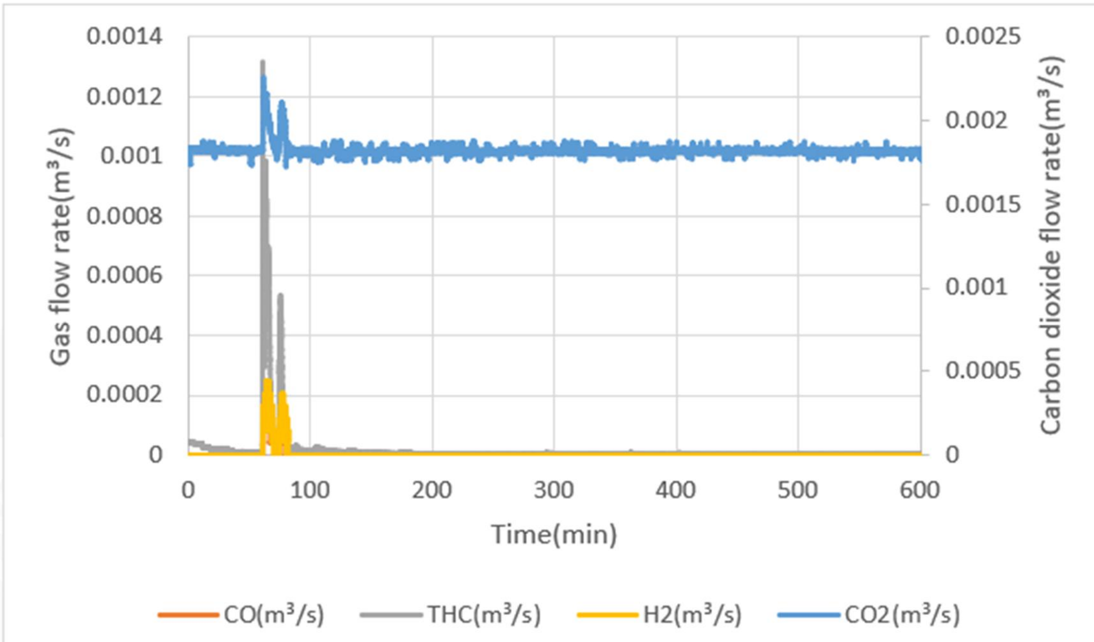
Attachment 6 - Gas generation graphs



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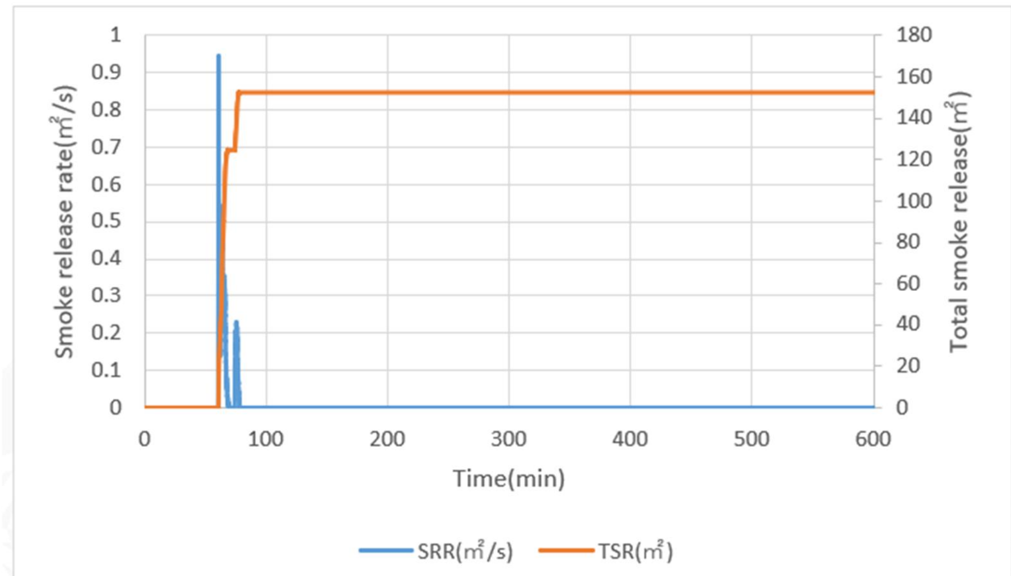
Attachment 7 - Smoke release graph



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Attachment 8 - Heat flux graph

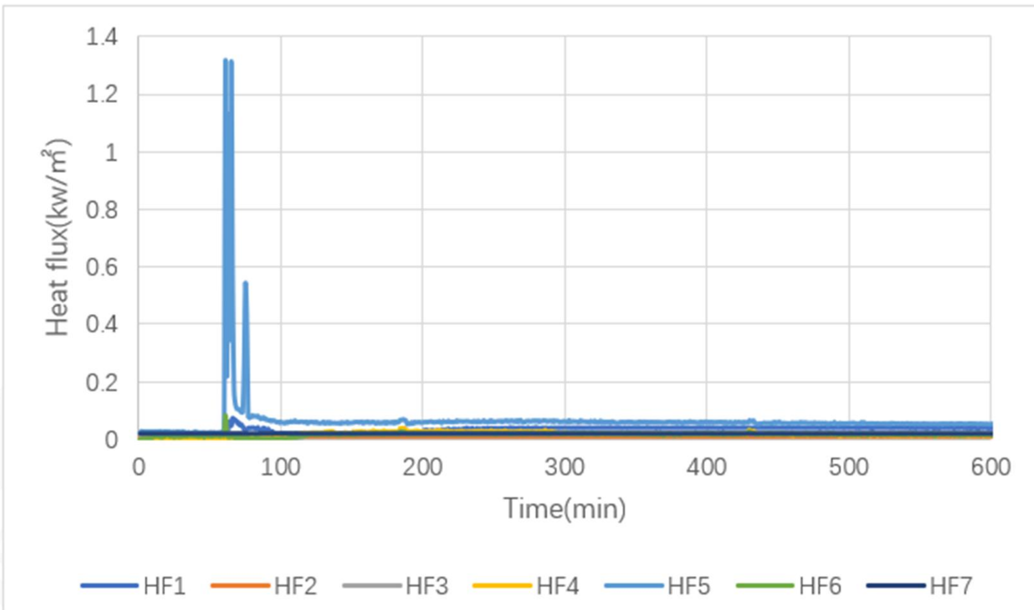

Table 6 – Maximum Heatflux measurement		
Location	**Heatflux limit (kW/m^2)**	**Measured maximum Heatflux (kW/m^2)**
HF1: Heat flux at module 3(Target unit 3), face to initial module	N/A	0.07
HF2: Heat flux at module 4(Target unit 2), face to initial module	N/A	0.01
HF3: Heat flux at Instrument wall A, the height of module 3(initial unit)	N/A	0.02
HF4: Heat flux at Instrument wall A, the height of module 4(Target unit 1)	N/A	0.03
HF5: Heat flux at Instrument wall B, the height of module 3(initial unit)	N/A	1.31
HF6: Heat flux at Instrument wall B, the height of module 4(Target unit 1)	N/A	0.08
HF7: Heat flux at egress path center, the height of initial module	1.3	0.02

Attachment 9 - Notable observation during test

Observation	**Time from test start (HH:MM:SS)**	**Comment**
Test start	14:45:08	Two pieces heater film on cell 33 was energized with thermal ramp 4.5 degree/min.
Venting	15:27:12	Cell 33 vent with pop sounds
Thermal runaway	15:45:25	Cell 33 Thermal runaway with smoke released and two pieces beater film on cell 33 all deenergized
Venting	15:46:44	Cell 32 vent with pop sounds
Thermal runaway	15:49:13	Cell 32 Thermal runaway with smoke released
Venting	15:48:12	Cell 34 vent with pop sounds
Thermal runaway	15:59:27	Cell 34 Thermal runaway with smoke released
Venting	16:02:04	pop sounds
Venting	16:56:07	pop sounds
Test end	8:00:00(the next day)	Data acquisition stopped

End of Report….